

PRIVACY-PRESERVING METHODS FOR DATA SCIENCE AND DISTRIBUTED SYSTEMS

EXAMPLES FOR PROJECT TOPICS FROM PREVIOUS YEARS

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PREDICTING ACADEMIC SUCCESS – ANALYSIS OF DIFFERENTIALLY PRIVATE MACHINE LEARNING MODELS AND DIFFERENT PRIVACY ACCOUNTANTS

Multiple factors can influence academic success. It can therefore be helpful to analyse such data to predict the expected success of a student and take according action. But such data includes sensitive information that we do not want to feed to a machine learning model in a non-private way. Accordingly, we aim to analyse such data in a differentially private manner, by training a machine learning model on an open source dataset [1], using DP-SGD. First we train a native model without any differential privacy mechanisms to achieve a baseline for future measurements. Then we train models using three different accounting methods: The Moments Accountant [2] which aggregates privacy costs in a Rényi DP framework, GDP Accounting [3] that uses a new approach to quantify privacy in the context of the Gaussian normal distribution, but can be converted to (ϵ, δ) -DP and lastly PRV Accounting [4] that exploits Fast Fourier Transform to add up privacy curves defined by Privacy Loss Random Variables. By comparing privacy estimations gained from different accounting methods, accuracy and runtime performance behaviour this project aims to find out if we can predict academic success using machine learning models to enable education institutions to offer assistance to students in need. We aim to choose optimal parameters for the developed models while keeping actual student data private and therefore strive to achieve a perfect trade-off between privacy and utility for this real-world example. This project was carried out as part of the lecture *Privacy-Preserving Methods for Data Science and Distributed Systems* at the University of Basel. The implementation and report were collaboratively developed by both authors through regular discourse, leading to the presented findings.

EVALUATING PRIVACY LEAKAGE USING MEMBERSHIP INFERENCE ATTACKS ON DIFFERENTIALLY PRIVATE MODELS

Machine learning models, especially neural networks, are increasingly deployed in sensitive applications involving health, finance, and user behavior. However, these models are prone to privacy vulnerabilities such as Membership Inference Attacks (MIA), where an adversary attempts to determine whether a particular data point was part of the model's training set. This project explores and implements multiple MIA techniques targeting three types of neural network models: a non-private baseline, a differentially private model trained with Opacus, and another trained with Pyvacy. Through both shadow model (white-box) and entropy-based (black-box) approaches, we empirically evaluate the susceptibility of these models to privacy leakage. Our experiments highlight that while differential privacy significantly mitigates exposure, non-private models remain highly vulnerable, validating the need for privacy-preserving training methods.

In this work, we evaluated the trade-offs involved in training a compact CNN with differentially private stochastic gradient descent (DP-SGD) for blood cell image classification on the BloodMNIST dataset. Although the final private model achieved only moderate accuracy, it was trained with differential privacy guarantees protecting against various privacy attacks on the training dataset. The model could be safely released — provided that training is performed once with $(2, 7 \cdot 10^{-9})$ -DP-SGD without additional hyperparameter tuning — to support ensemble methods or transfer learning for data-scarce applications. Our experiments also showed that the additional privacy cost of hyperparameter search resulted in only a negligible gain in accuracy. We hope our findings contribute to tuning the utility–privacy trade-off in medical image classification with DP-SGD in practice.

Recommender systems are algorithms designed to suggest relevant items to users based on their preferences, behaviors, or specified constraints. They are widely used in domains such as e-commerce, streaming services, and social media platforms. Because recommender systems require extensive user data to provide personalized recommendations, they raise significant privacy concerns. In the past, attacks have demonstrated how recommender systems can leak sensitive information, such as a user's preferences, demographic data, or even private interactions [8]. In this project, we aim to analyze privacy-preserving recommender systems for movies that use differentially private training to protect user data, using the MovieLens 1M dataset [9].

Differential privacy (DP) offers a framework to ensure that the inclusion or exclusion of a single data point does not significantly affect the outcome of any analysis. Thus it is protecting the privacy of each individual in the dataset. Many differential privacy mechanisms use a centralized algorithm. Sensitive data of different users is collected on a server where the mechanism is applied. This is typically achieved by adding random noise to the published values in order to make them differentially private. An example for such an approach is the *Laplace mechanism*. However, centralized frameworks come with certain disadvantages. The server has unrestricted access to the confidential data of individual users. If the server wants to harm its users or simply handles the data carelessly, significant leaks of private information may occur.

Local DP (LDP) algorithms attempt to solve this problem by obscuring the data before sending it to the server. In this project we concentrate on the *Count Mean Sketch* algorithm, a local DP technique proposed and deployed on a large scale by Apple. We compare the algorithm with other local DP frameworks such as *Randomized Response* and *Histogram Encoding*. Furthermore, we analyse how these local frameworks perform in relation to the Laplace mechanism. Our evaluation covers privacy, utility and performance metrics.

ANALYSIS OF THE NHEFS DATASET SIMULATING A DIFFERENTIALLY PRIVATE DATABASE

This report presents an analysis of the NHEFS (National Health and Nutrition Examination Survey Epidemiologic Follow-up Study) dataset simulating the usage of a differentially private database. The NHEFS dataset is a rich longitudinal study that tracks the health and nutritional status of individuals over time. This analysis aims to draw insights from how various factors such as education, marital status, and income influence health outcomes, and especially to understand the causal effects of lifestyle choices like smoking cessation on subsequent health changes.

Differential privacy is employed in this analysis to ensure that the privacy of individuals in the dataset is preserved. Differential privacy provides a mathematical guarantee that the inclusion or exclusion of a single individual's data does not significantly affect the results of the analysis, thereby protecting individual privacy. By utilizing differential privacy, we can perform meaningful analysis on the dataset while adhering to strict privacy standards.